

8.0

THERMOCHEMISTRY

8.1 Concept of enthalpy

Learning objectives

- a) explain endothermic and exothermic reactions using the energy profile diagrams
- b) State standard conditions of reaction and define the following terms
 - i. Enthalpy
 - ii. Standard Enthalpy
- c) Define enthalpy of
 - i. formation
 - ii. combustion
 - iii. atomisation
 - iv. neutralisation
 - v. hydration
 - vi. solution (dissolution)
- d) Write thermochemical equation for each enthalpy

Terms (must remember to avoid confusion!)

- **Heat** = energy transferred between two bodies of different temperatures
- **System** = any specific part of the universe
- **Surroundings** = everything that lies outside the system
- **Open system** = a system that can exchange mass and energy with its surroundings
- **Closed system** = a system that allows the exchange of energy with its surroundings
- **Isolated system** = a system that does not allow the exchange of either mass or energy with its surroundings
- **Energy** = the ability to do work
 - SI unit of energy is **kg m² s⁻²** or **Joule** (J)
 - Non SI unit of energy is **calorie** (Cal)
 - 1 Cal = 4.184 J

Enthalpy, ΔH

- The heat content of the reacting system is known as enthalpy of the system.
- The absolute value of the enthalpy (H) of a system cannot be determined. We can only determine the **change in enthalpy, ΔH** .

$$\Delta H = H_{(products)} - H_{(reactants)}$$

Types of reaction

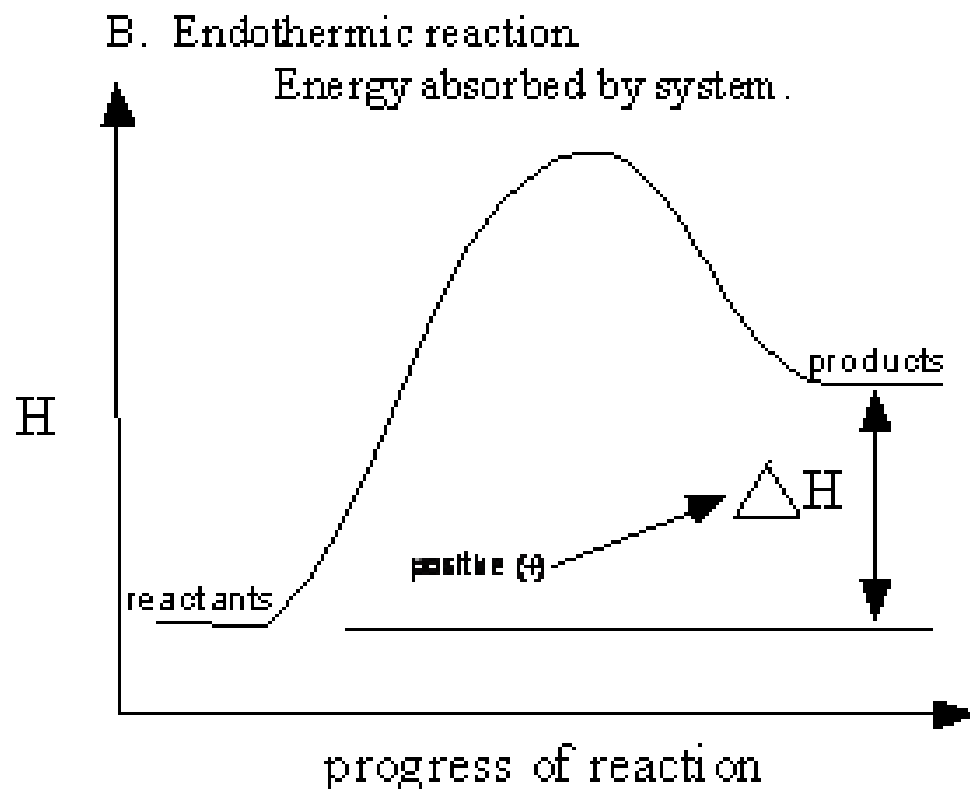
Endothermic reaction

- enthalpy of products $>$ enthalpy of reactants
- ΔH is positive
- Heat is absorbed from the surroundings

Exothermic reaction

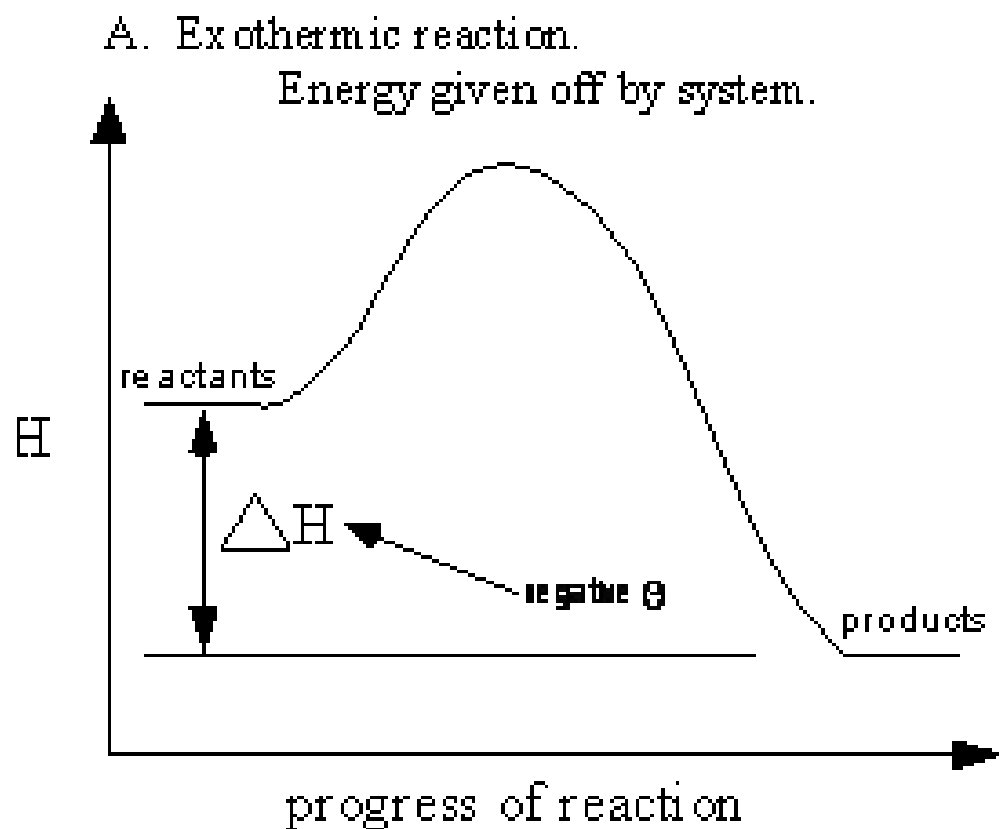
- enthalpy of products $>$ enthalpy of reactants
- ΔH is negative
- Heat is released from the surroundings

Endothermic reaction



Energy diagram for endothermic reaction

Exothermic reaction



Energy diagram for exothermic reaction

Examples of endothermic and exothermic reactions

Endothermic reaction

- melting ice cubes
- reacting ethanoic acid with sodium carbonate

Exothermic reaction

- burning of propane
- freezing water

- **Enthalpy of reaction:**

- The enthalpy change associated with a chemical reaction.

- **Standard enthalpy, ΔH°**

- The enthalpy change for a particular reaction that occurs at 298K and 1 atm (standard state)

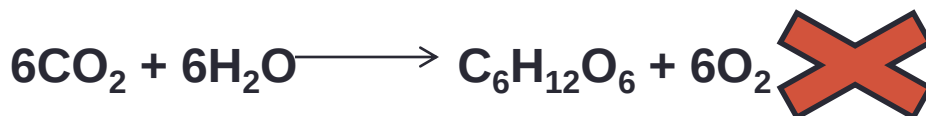
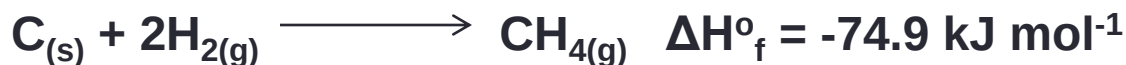
Types of enthalpy

We will learn several types of enthalpies:

- ☐ Enthalpy of formation
- ☐ Enthalpy of combustion
- ☐ Enthalpy of atomisation
- ☐ Enthalpy of neutralisation
- ☐ Enthalpy of hydration
- ☐ Enthalpy of solution(dissolution)

Enthalpy of formation, ΔH_f

- The heat change when **1 mole of compound** is formed from its **constituent elements** with both the compound and elements in their standard state
 - It may be endothermic or exothermic
 - Any accompanying equation should involve the formation of 1 mole of the compound



- The standard enthalpy of formation for an element in its standard state is ZERO!!!! Elements in their standard state are not formed, they just are. So, ΔH°_f for C (s, graphite) is zero, but the ΔH°_f for C (s, diamond) is 2 kJ/mol. That is because graphite is the standard state for carbon, not diamond.



Enthalpy of combustion, ΔH_c

- The heat released when **1 mole of substance** is burned completely in **excess oxygen**.
- You should ensure just 1 mole of substance appears in the equation to accompany the delta H value which is always negative i.e. **always exothermic**.



Enthalpy of atomisation, ΔH_a

- The heat absorbed when one mole of free gaseous atoms is formed from its element.
- ΔH_a is always positive because it involves only breaking of bonds



Enthalpy of neutralisation, ΔH_n

- The heat released when 1 mole of water, H_2O is formed from the neutralization of acid and base.
- Enthalpy changes of neutralisation are always negative - heat is given out when an acid and an alkali react.



Enthalpy of hydration, ΔH_{hyd}

- The hydration enthalpy is the heat released when **1 mole of gaseous ions** dissolve in sufficient water to give **an infinitely dilute solution**.
- Hydration enthalpies are **always negative**.



Enthalpy of solution, ΔH_{soln}

- The heat change when 1 mole of an ionic substance dissolves in water to give a solution of infinite dilution.



THERMOCHEMISTRY

8.2 CALORIMETRY

Learning objectives

- a) Define
 - i. Heat capacity, C
 - ii. Specific heat capacity, c
- b) Calculate heat change in a
 - i. Constant-Pressure Calorimetry (simple calorimeter)
 - ii. Constant-Volume Calorimetry (bomb calorimeter)

CALORIMETRY

- **Calorimetry**, derived from the Latin *calor* meaning heat, and the Greek *metry* meaning to measure, is the method in science to measure the amount of heat
- A **calorimeter** is a device used to measure heat of reaction
 - i. Simple calorimeter
 - ii. Bomb calorimeter

Before that.....

- You need to know what is heat capacity, C and specific heat capacity, c !!!!
- If you cannot grab the idea behind these two terms, then everything will look like $@£\%^{\wedge}&*()*\&\%^{\circ}\$$ to you!!!

Heat capacity, C

- It is the amount of heat needed to raise the temperature of a given quantity of substance by 1 °C

$$q = C\Delta T$$

C = molar heat capacity of substance ($\text{J } ^\circ\text{C}^{-1}$)

ΔT = change in temperature($^\circ\text{C}$)

$$= T_{\text{final}} - T_{\text{initial}}$$

- Unit = **J**

Specific heat capacity, c

- It is the amount of heat needed to raise the temperature of **1 gram** of any substance by 1 °C

$$q = mc\Delta T$$

m = mass(g)

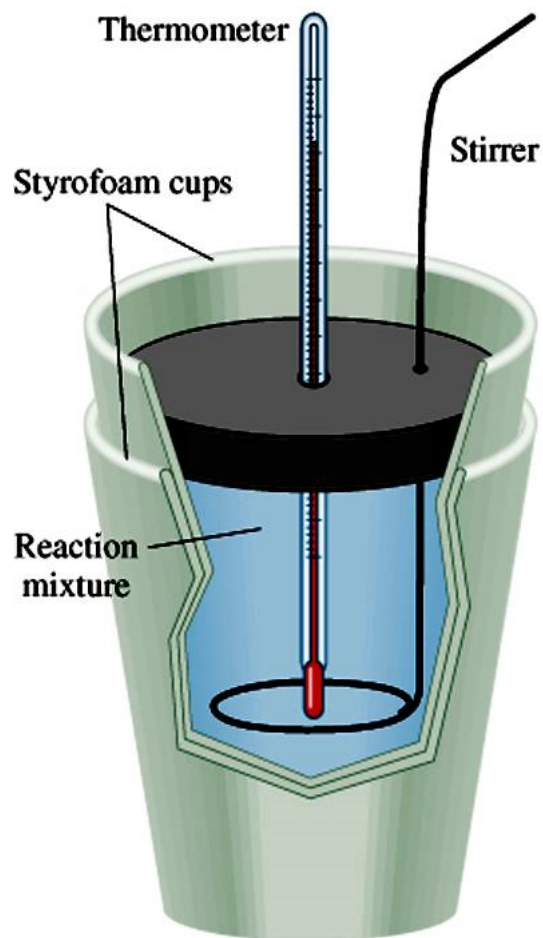
C = specific heat capacity of substance ($\text{J g}^{-1} \text{ } ^\circ\text{C}^{-1}$)

ΔT = change in temperature ($^\circ\text{C}$)

$$= T_{\text{final}} - T_{\text{initial}}$$

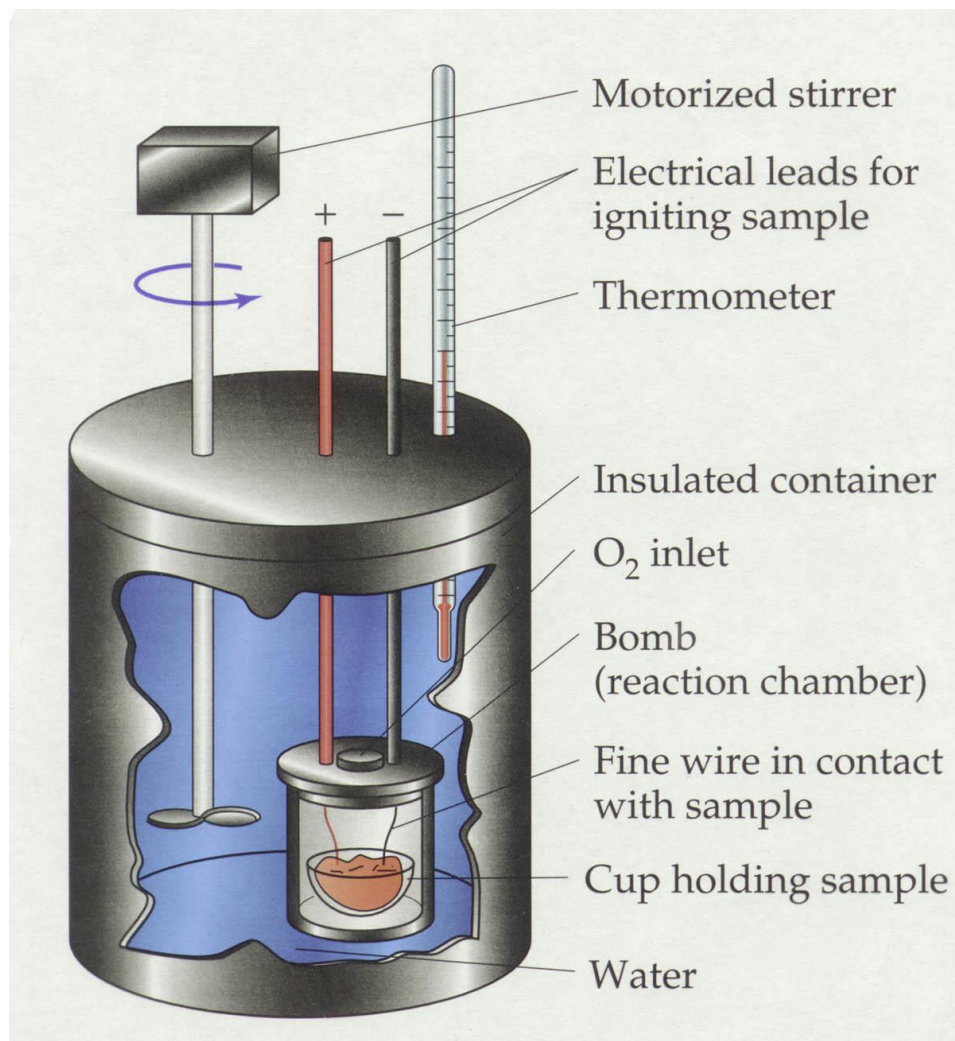
- Unit = J

Simple Calorimeter(Constant-pressure Calorimetry)



- Two stacked Styrofoam cup
- When a reaction occurs, the heat evolved is absorbed by the solution and thermometer
- It is assumed that no heat is lost due to insulation by the two cups

Bomb Calorimeter (Constant-volume Calorimetry)



- Very precise measurement of heat of reaction
- Used to measure heat change for reactions that needs to be initiated; combustion
- The heat is absorbed by the bomb, the water and any piece of equipment sticking into the water

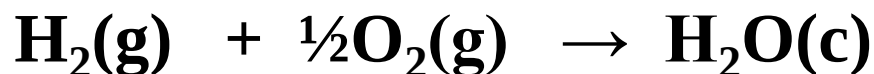
Example 1

In an experiment, 0.100 g of H_2 and excess of O_2 were compressed into a 1.00 L bomb and placed into a calorimeter with **heat capacity** of $9.08 \times 10^4 \text{ J}^\circ\text{C}^{-1}$. The initial temperature of the calorimeter was 25.000°C and finally it increased to 25.155°C . Calculate the amount of heat released in the reaction to form H_2O , expressed in kJ per mole.

Solution

Heat released = Heat absorbed by the calorimeter

$$\begin{aligned} q &= C\Delta T \\ &= (9.08 \times 10^4 \text{ J}^\circ\text{C}^{-1}) \times (0.155^\circ\text{C}) \\ &= 1.41 \times 10^4 \text{ J} \\ &= 14.1 \text{ kJ} \end{aligned}$$



$$\begin{aligned} \text{mole of H}_2 &= \frac{0.100}{2.016} \\ &= 0.0496 \text{ mol} \end{aligned}$$

moles of H₂O = mole of H₂

0.0496 mol of H₂O released 14.1 kJ energy

$$\text{1 mol H}_2\text{O released} = \frac{14.1}{0.0496} \text{ kJ}$$

$$= 284 \text{ kJ}$$

$\Delta\text{Heat of reaction, } \Delta H = - 284 \text{ kJ mol}^{-1}$

Example 2

Calculate the amount of **heat released** in a reaction in an **aluminum calorimeter** with a mass of **3087.0 g** and contains **1700.0 mL** of water. The initial temperature of the calorimeter is **25.0°C** and it increased to **27.8°C**.

Given:

Specific heat capacity of aluminum = $0.553 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$

Specific heat capacity of water = $4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}$

Water density = 1.0 g mL^{-1}

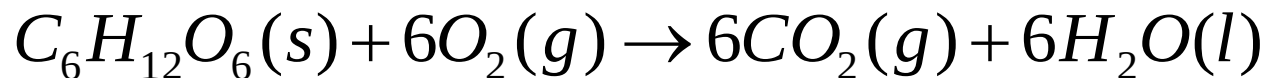
$\Delta T = (27.8 - 25.0) ^{\circ}\text{C} = 2.8 ^{\circ}\text{C}$

Solution

Heat released = Heat absorbed by aluminium calorimeter + Heat absorbed by water

$$\begin{aligned} q &= m_w c_w \Delta T + m_c c_c \Delta T \\ &= (1700.0 \text{ g})(4.18 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1})(2.8 \text{ }^{\circ}\text{C}) + \\ &\quad (3087.0 \text{ g})(0.553 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1})(2.8 \text{ }^{\circ}\text{C}) \\ &= 24676.71 \text{ J} \\ &= 24.7 \text{ kJ} \end{aligned}$$

Example 3



The burning of 2.06 g of glucose in a bomb calorimeter causes the temperature of the water to increase from 29.45 °C to 31.44 °C.

The calorimeter contains 982.0 g of water. The heat capacity of the calorimeter is 814 J/°C. The specific heat of water is 4.184 J/g °C

- i. Calculate the heat of combustion of 1 mol of glucose.
- ii. If you have a dietary supplement of 1200 Dietary Calories per day, what mass of glucose would supply this energy? (1 Dietary Calorie = 4.184 kJ)

Solution

i. Bomb calorimeter = constant volume

$$\Delta T = (31.44 - 29.45)^\circ\text{C} = 1.99^\circ\text{C}$$

Heat released = Heat absorbed by calorimeter + Heat absorbed by water

$$q = mc\Delta T + C_{\text{cal}}\Delta T$$

where c is the specific heat of water and C_{cal} is the heat capacity of the calorimeter

$$\begin{aligned} q &= (982 \text{ g} \times 4.184 \text{ J}/^\circ\text{C} \times 1.99^\circ\text{C}) + (814 \text{ J}/^\circ\text{C} \times 1.99^\circ\text{C}) \\ &= 8176.29 \text{ J} + 1619.86 \text{ J} \\ &= 9.796 \times 10^3 \text{ J} \end{aligned}$$

Combustions are exothermic; signs are negative

$$q = -9.796 \times 10^3 \text{ J for } 2.06 \text{ g of glucose burned}$$

$$q = \frac{-9.796 \times 10^3 \text{ J}}{2.06 \text{ g}} \times \frac{180.16 \text{ g}}{1 \text{ mol}}$$

$$q = -8.567 \times 10^5 \text{ J mol}^{-1}$$

$$q = -857 \text{ kJ mol}^{-1}$$

$$\begin{aligned} \text{ii) Mass of glucose} &= 1200 \times \frac{4.184 \text{ kJ}}{1} \times \frac{1 \text{ mol}}{856.7 \text{ kJ}} \times \frac{180.16 \text{ g}}{1 \text{ mol}} \\ &= 1055.85 \text{ g} \end{aligned}$$



THERMOCHEMISTRY

8.3 Hess's Law

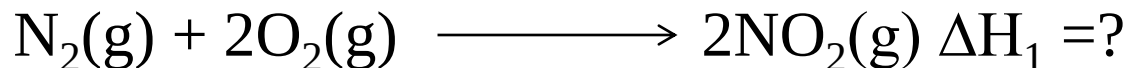


LEARNING OBJECTIVES

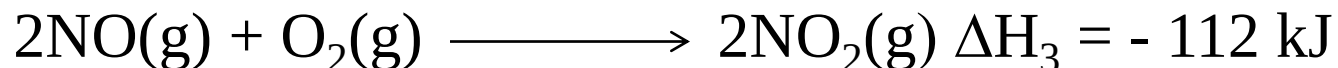
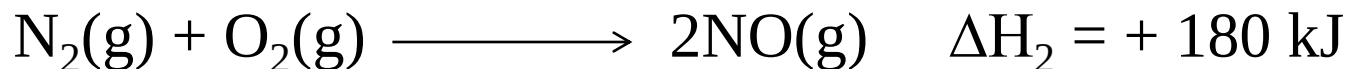
- State Hess's law
- Apply Hess's law to calculate enthalpy changes using the algebraic method and the energy cycle method

HESS'S LAW

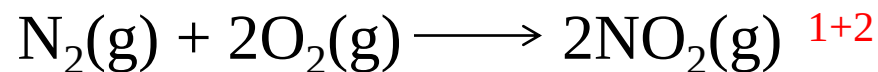
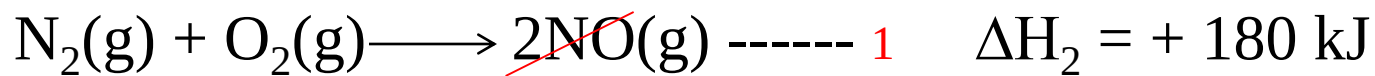
- Hess's law states that **total enthalpy change for a reaction is the same whether it occurs in one step or in many steps.**
- Example: To determine ΔH_1 for the following reaction:



given



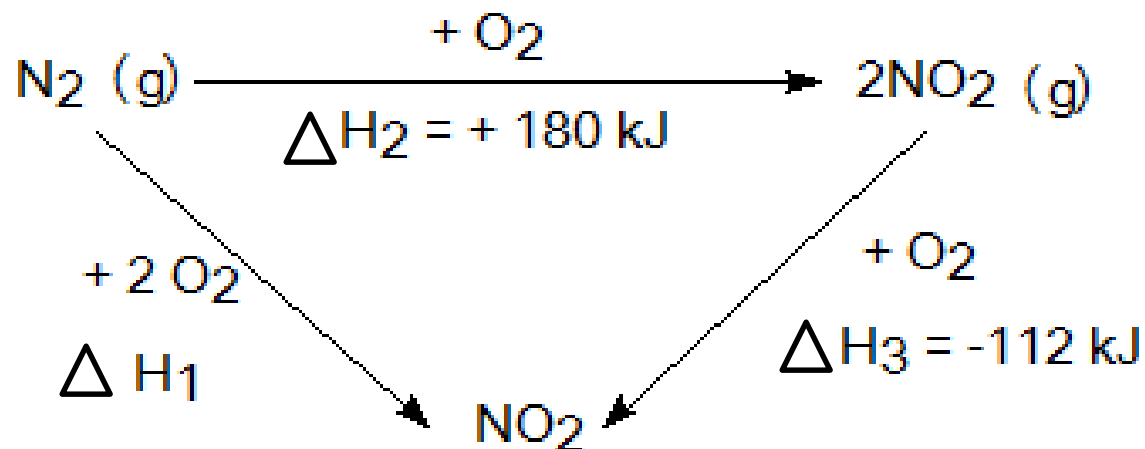
ALGEBRAIC METHOD



$$\Delta H_1 = \Delta H_2 + \Delta H_3 = (180 - 112) \text{ kJ} = + 68 \text{ kJ}$$

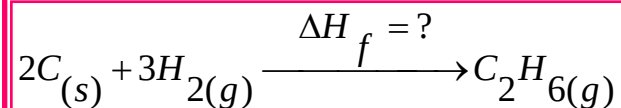
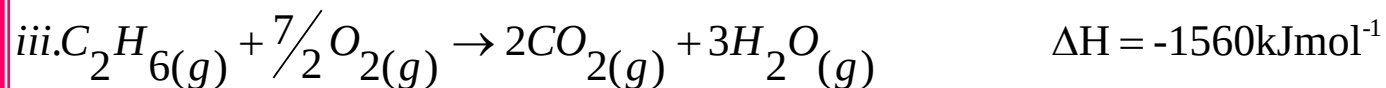
**BE CAREFUL WITH THE CANCELLATION!!!
MAKE SURE YOU GET THE RIGHT
PRODUCT.....**

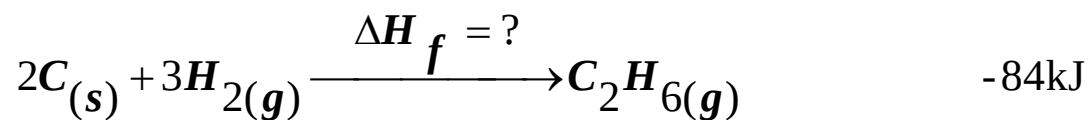
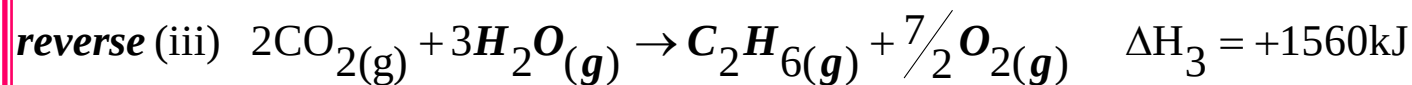
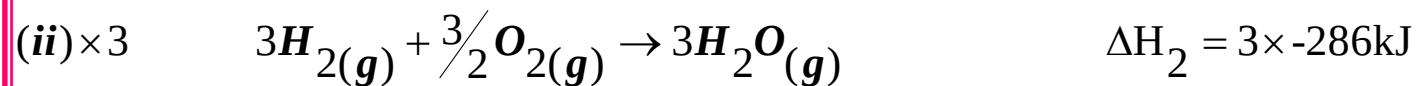
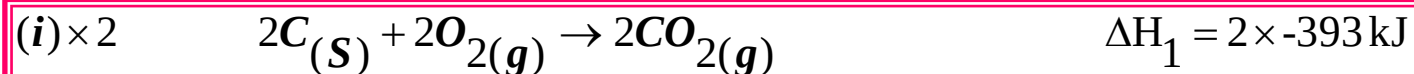
ENERGY CYCLE METHOD



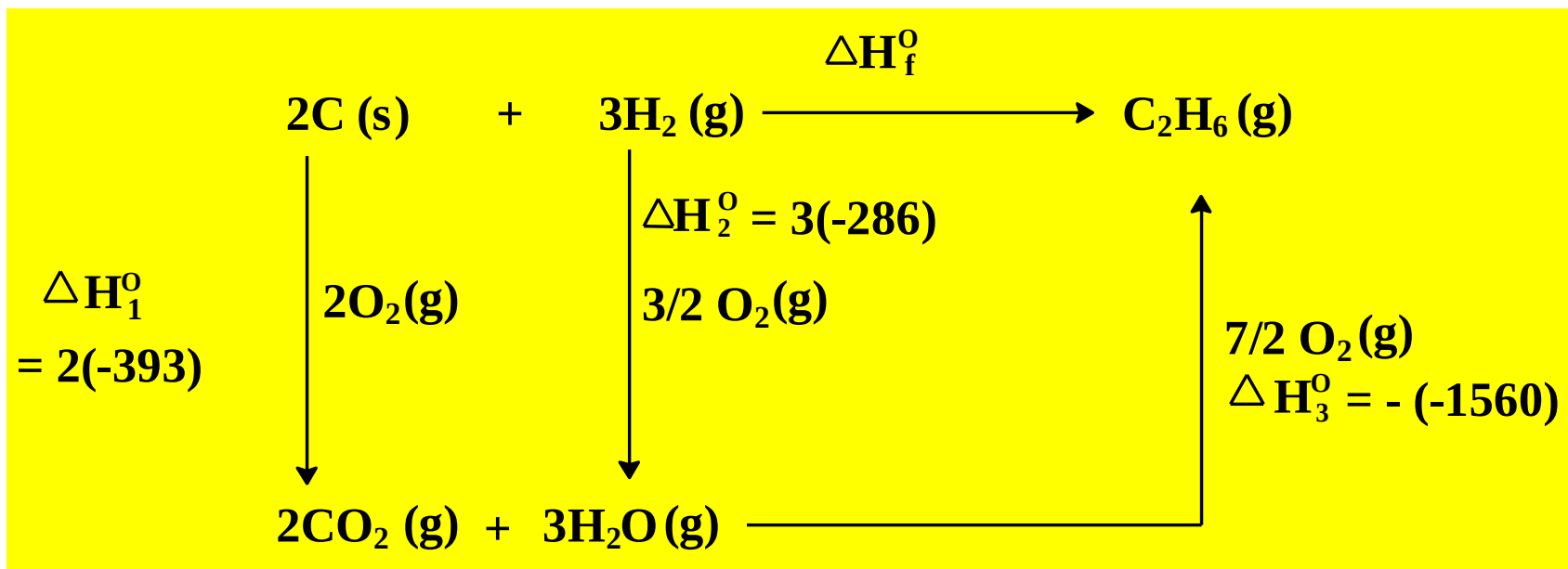
$$\begin{aligned}\Delta H_1 &= \Delta H_2 + \Delta H_3 \\ &= +180 - 112 \\ &= +68 \text{ kJ}\end{aligned}$$

Example 1





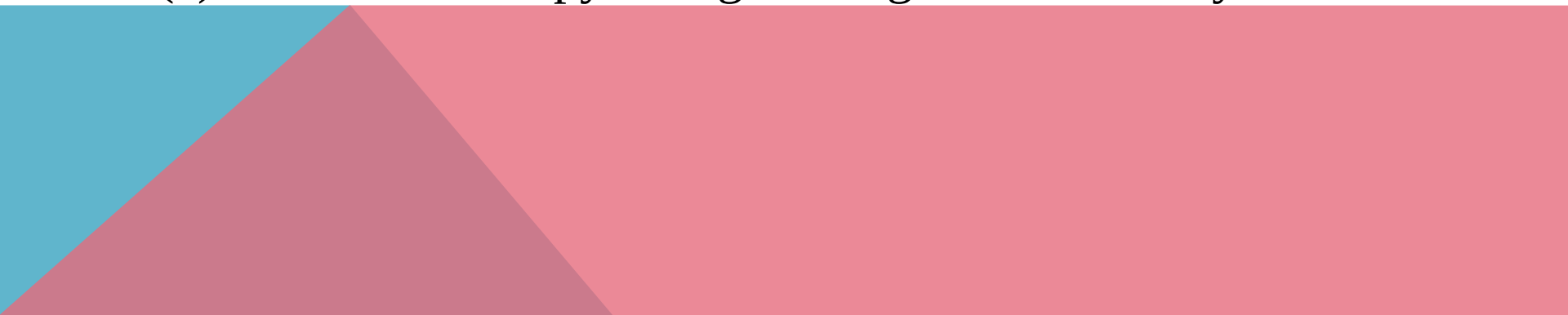
$$\begin{aligned} \Delta H_f &= \Delta H_1 + \Delta H_2 + \Delta H_3 \\ &= -84 \text{ kJ} \end{aligned}$$



THERMOCHEMISTRY

BORN – HABER CYCLE

LEARNING OBJECTIVES

- (a) Define lattice energy and electron affinity
 - (b) Explain the following effects on the magnitude of the lattice energy
 - i. ionic charges
 - ii. Ionic radii
 - (a) Explain the dissolution process of ionic solids.
 - (b) Construct Born-Haber cycle for simple ionic solids using energy cycle diagram.
 - (c) Calculate enthalpy changes using Born-Haber cycle
- 

ELECTRON AFFINITY

- Electron affinity is the enthalpy change when 1 mole of gaseous atoms gains 1 mole of electrons to form 1 mole of gaseous ions
- For most atoms, energy is released when electron is added, EA usually will have a negative sign



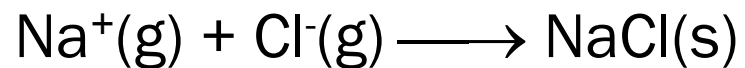
LATTICE ENERGY

- The lattice formation enthalpy is defined as the energy released when one mole of the compound is formed from its gaseous ions. Lattice formation enthalpies are always negative.
 - The lattice dissociation enthalpy is defined as the energy required to completely separate one mole of a solid (ionic compound) into gaseous ions. Lattice dissociation enthalpies are always positive.
- 



$$\Delta H_{\text{lattice}} = +771 \text{ kJ mol}^{-1}$$

(lattice dissociation)



$$\Delta H_{\text{lattice}} = -771 \text{ kJ mol}^{-1}$$

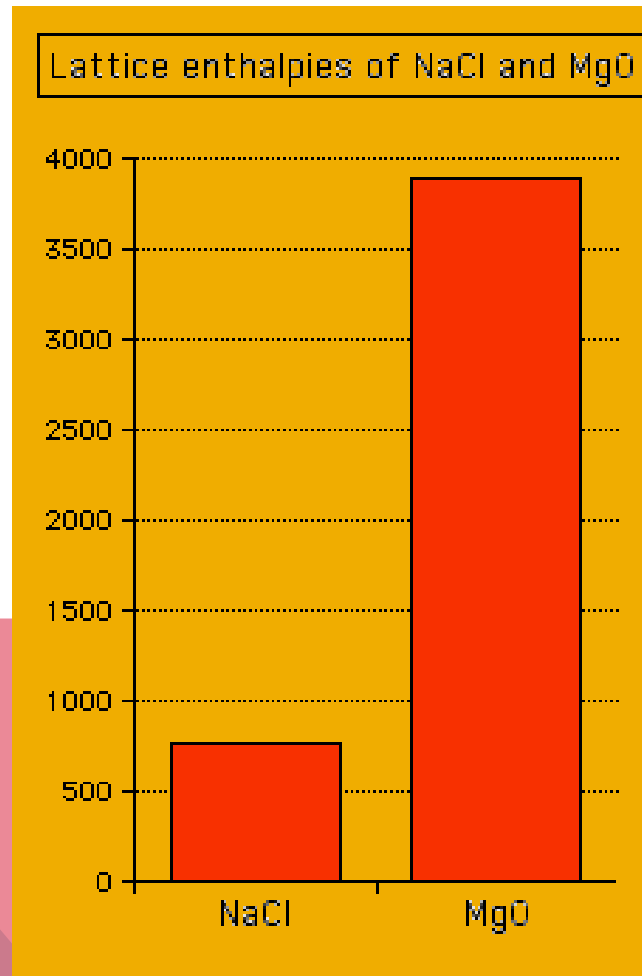
(lattice formation)

FACTORS AFFECTING LATTICE ENTHALPY

Charges on ions

LE increases as the charges increases

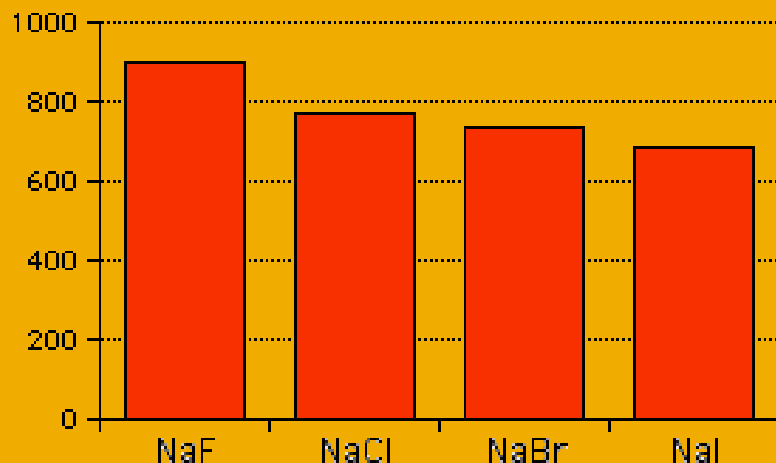
LE $\text{MgO} > \text{NaCl}$;
Mg has +2
charge, while has
+1 charge



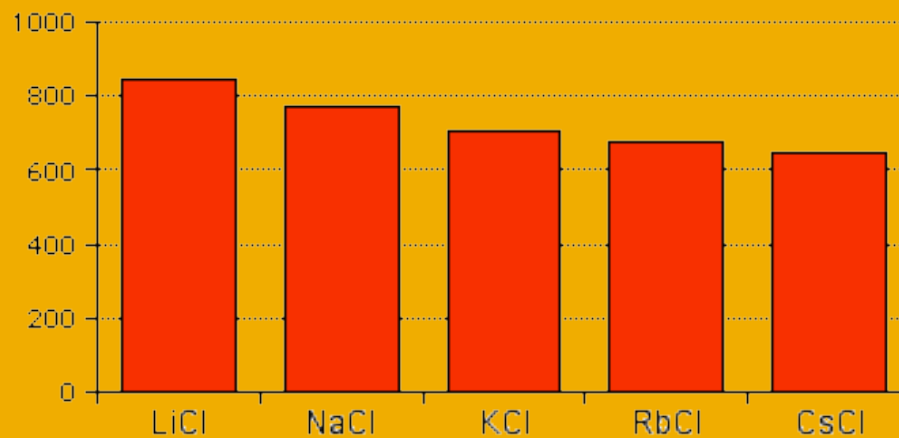
The radius of the ions

- The smaller the radius, the stronger the lattice enthalpy
- Ions are closer together in the lattice, increases strength of attractions

Lattice enthalpies of sodium halides



Lattice enthalpies of Group 1 chlorides



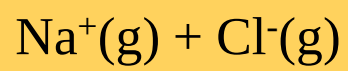
DISSOLUTION PROCESS OF IONIC SOLIDS

- Dissolution is the process of dissolving solid using suitable solvent
- The process is split up into:
 - A step of breaking up the ionic lattice into gaseous ions
 - Hydrating the ions by surrounding them with polar water molecules

EXAMPLE

- Breaking up ionic lattice (NaCl) into gaseous phase Na^+ and Cl^- requires a lot of energy ($\Delta H_{\text{lattice}}$)
- When these isolated gaseous ions are placed in water, they become hydrated due to the ion-dipole forces of attraction (ΔH_{hydr})

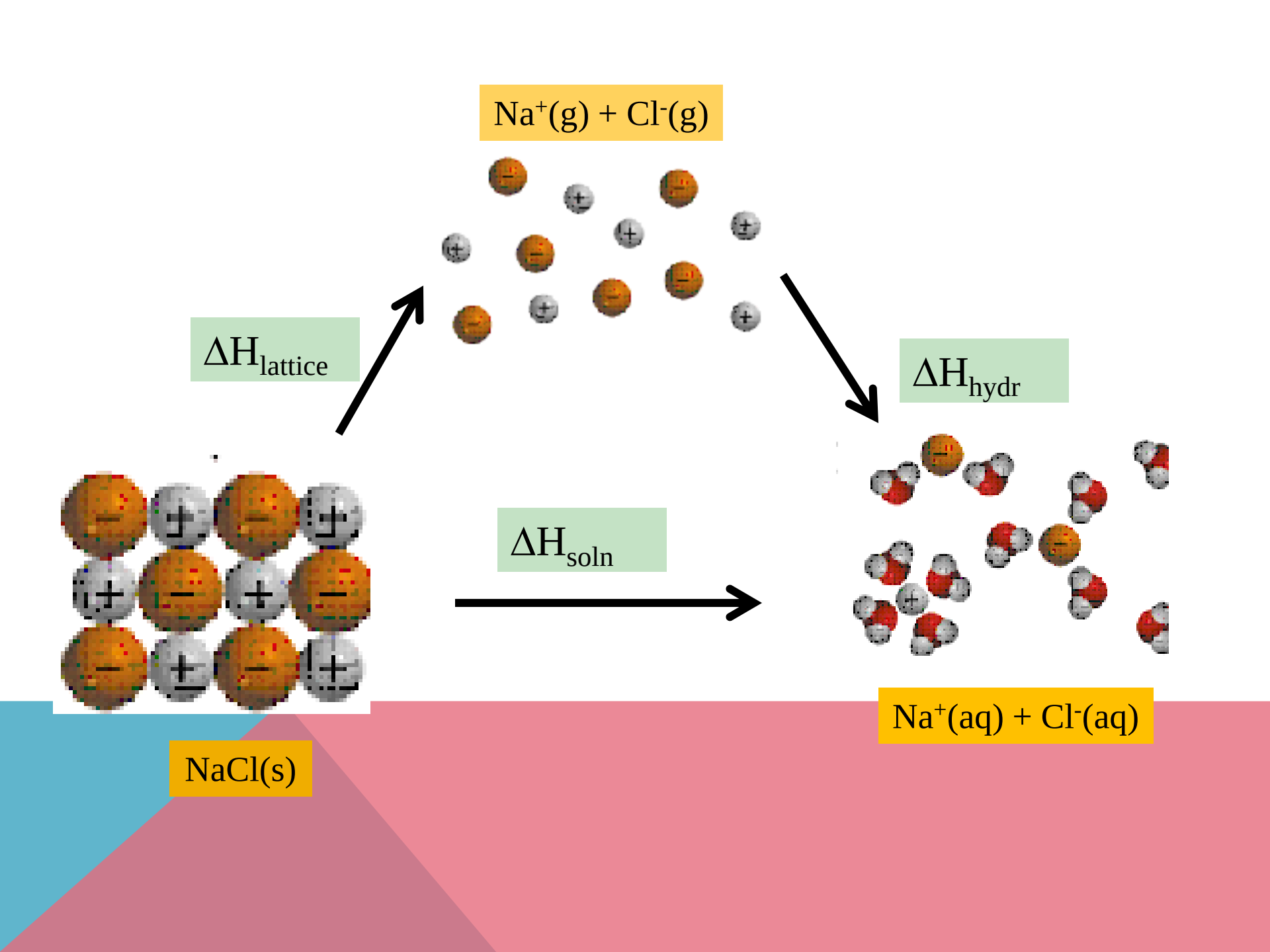
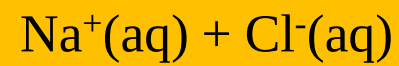
$$\Delta H_{\text{soln}} = -\Delta H_{\text{lattice}} + \Delta H_{\text{hydr}}$$



$\Delta H_{\text{lattice}}$

ΔH_{hydr}

ΔH_{soln}



BORN HABER CYCLE

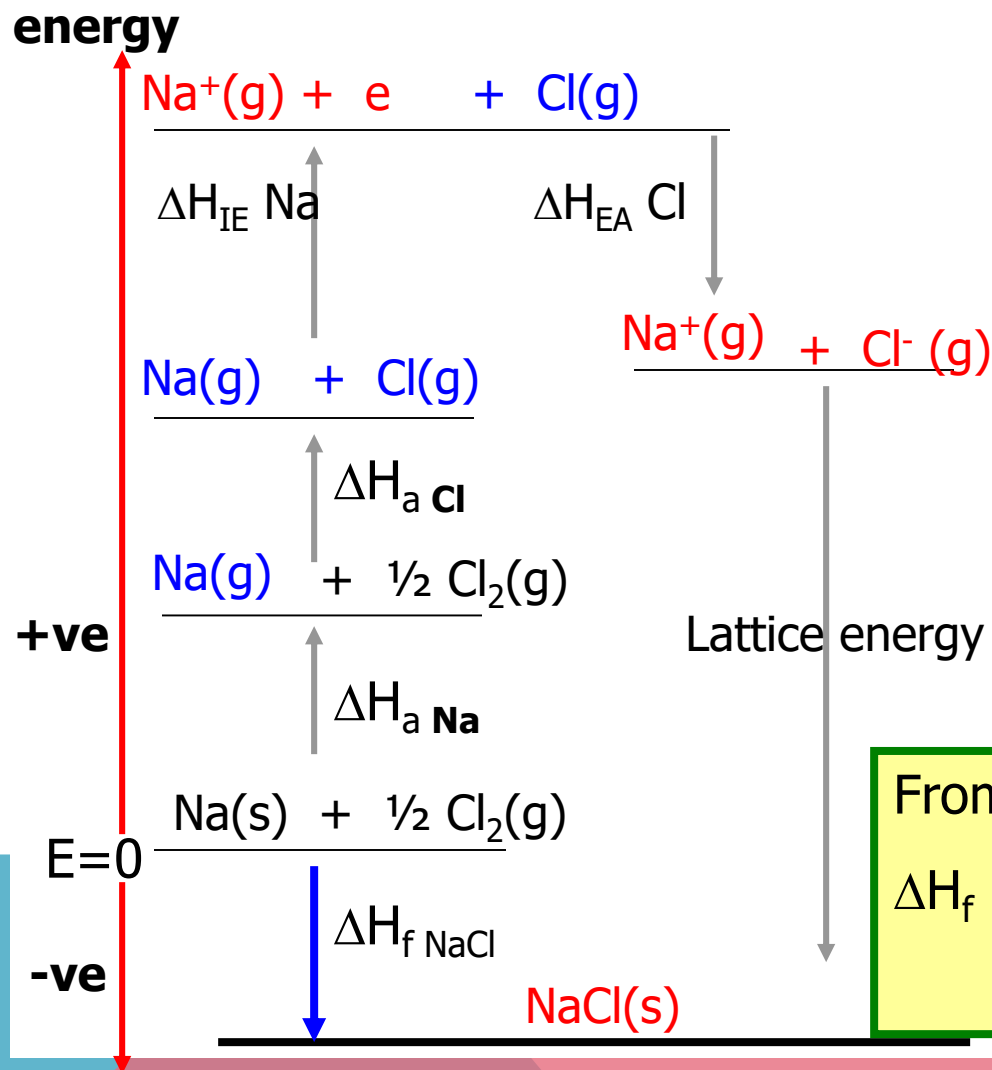
- The process of ionic bond formation occurs in a few stages. At each stage the enthalpy changes are considered.
- The Born Haber cycle is often used to calculate the lattice energy of an ionic compound.
- In the Born-Haber cycle energy diagram, by convention, positive values are denoted as going upwards, negative values as going downwards.

EXAMPLE

Given;

- i. Enthalpy of formation NaCl = -411 kJmol^{-1}
- ii. Enthalpy of atomization of Na = $+108 \text{ kJmol}^{-1}$
- iii. First ionization energy of Na = $+500 \text{ kJmol}^{-1}$
- iv. Enthalpy of atomization of Cl = $+122 \text{ kJmol}^{-1}$
- v. Electron affinity of Cl = -364 kJmol^{-1}
- vi. Lattice energy of NaCl = ?

Example: A Born-Haber cycle for NaCl



From Hess's Law:

$$\Delta H_{f \text{ NaCl}} = \Delta H_{a \text{ Na}} + \Delta H_{a \text{ Cl}} + \text{IE}_{\text{Na}} + \text{EA}_{\text{Cl}} + \text{Lattice Energy}$$

CALCULATION:

$$\Delta H_f^0 = \Delta H_S + IE + \Delta H_{a(Cl)} + EA + \Delta H_{lattice}$$

$$\Delta H_{lattice} = \Delta H_f^0 - [\Delta H_{a(Na)} + IE + \Delta H_{a(Cl)} + EA]$$

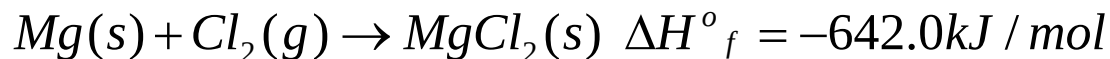
$$\Delta H_{lattice} = -411\text{kJ} - [+108\text{kJ} + 500\text{kJ} + 122\text{kJ} + (-364\text{kJ})]$$

$$\Delta H_{lattice} = -777\text{kJ}$$

EXERCISE

- Calculate the enthalpy of formation NaCl_2 . Use the data below:
 - i. Enthalpy of atomization of sodium = $+108 \text{ kJmol}^{-1}$
 - ii. First ionization energy of sodium = $+500 \text{ kJmol}^{-1}$
 - iii. Second ionization energy of sodium = $+4562 \text{ kJmol}^{-1}$
 - iv. Enthalpy of atomization of chlorine = $+121 \text{ kJmol}^{-1}$
 - v. Electron affinity of chlorine = -364 kJmol^{-1}
 - vi. Lattice energy of NaCl_2 = -2489 kJmol^{-1}

EXERCISE



- i. Enthalpy of atomisation of magnesium $\Delta H_a(\text{Mg}) = +149.0 \text{ kJ/mol}$
- ii. Enthalpy of atomisation of chlorine $\Delta H_a(\text{Cl}) = +243.0 \text{ kJ/mol}$
- iii. First ionisation ionisation enthalpy of magnesium $\Delta H_{\text{IE}}(1) = +737.7 \text{ kJ/mol}$
- iv. Second ionisation ionisation enthalpy of magnesium $\Delta H_{\text{IE}}(2) = +1450.7 \text{ kJ/mol}$
- v. Electron affinity of chlorine $\Delta H_{\text{EA}}(\text{Cl}) = -697.2 \text{ kJ/mol}$
- vi. Lattice energy of $\text{MgCl}_2 = ?$